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Answer

Okay, I can help you with this problem. This is a problem related to **Sets and Relations, specifically Equivalence Relations.** It's more of a **Conceptual** type of question. Here's a breakdown of how to solve it: **Understanding Equivalence Relations** First, let's recap what an equivalence relation is. A relation on a set A is an equivalence relation if it satisfies three properties: * **Reflexive:** For every element `a` in A, `(a, a)` is in the relation. * **Symmetric:** If `(a, b)` is in the relation, then `(b, a)` is also in the relation. * **Transitive:** If `(a, b)` and `(b, c)` are in the relation, then `(a, c)` is also in the relation. Equivalence relations partition the set into disjoint subsets called equivalence classes. **Solving the Problem** We need to find the number of *non-empty* equivalence relations on the set A = {1, 2, 3}. It's helpful to think about how equivalence relations create partitions of the set. Here's a systematic way to find the equivalence relations: 1. **All elements related:** The simplest equivalence relation is where *all* elements are related to each other. This means the relation contains (1,1), (2,2), (3,3), (1,2), (2,1), (1,3), (3,1), (2,3), (3,2). This corresponds to one equivalence class: {1, 2, 3}. There's only **1** such relation. 2. **Two equivalence classes:** We can partition the set into two equivalence classes. There are three ways to do this: * {1} and {2, 3}: This means 2 and 3 are related to each other, but 1 is only related to itself. The relation would contain (1,1), (2,2), (3,3), (2,3), (3,2). * {2} and {1, 3}: This means 1 and 3 are related to each other, but 2 is only related to itself. The relation would contain (1,1), (2,2), (3,3), (1,3), (3,1). * {3} and {1, 2}: This means 1 and 2 are related to each other, but 3 is only related to itself. The relation would contain (1,1), (2,2), (3,3), (1,2), (2,1). So, there are **3** such relations. 3. **Three equivalence classes:** Each element is only related to itself. This means the relation contains only (1,1), (2,2), and (3,3). This corresponds to three equivalence classes: {1}, {2}, and {3}. There's only **1** such relation. **Total Count** Adding up the number of equivalence relations from each case: 1 (one class) + 3 (two classes) + 1 (three classes) = 5 **Therefore, the number of non-empty equivalence relations on the set {1, 2, 3} is 5. So the correct answer is (3).** **Alternative method: Using Bell Numbers** Bell numbers count the number of partitions of a set. The *n*th Bell number, *B_n*, represents the number of ways to partition a set with *n* elements. * B₀ = 1 (empty set) * B₁ = 1 * B₂ = 2 * B₃ = 5 * B₄ = 15 ...and so on. In our case, we have a set with 3 elements, so we need to find B₃, which is 5. **Practice Problems & NCERT/PYQ References** 1. **NCERT:** Review the chapter on Relations and Functions in your Class 11 NCERT textbook. Pay close attention to the definitions of relations, equivalence relations, and partitions of sets. 2. **PYQs:** Solve previous year JEE questions related to relations and functions, focusing on problems involving equivalence relations. This will help you understand the types of questions asked and the level of difficulty. 3. **Practice Problems:** * Find the number of equivalence relations on the set {1, 2, 3, 4}. * Let R be a relation on the set of integers Z defined by (a, b) ∈ R if a - b is divisible by 5. Show that R is an equivalence relation. Find the equivalence classes. **Key Takeaways & Motivation** * Understanding the *definition* of an equivalence relation is crucial. * Thinking about how equivalence relations partition a set makes the problem easier to visualize. * Bell numbers provide a shortcut for counting partitions, but it's important to understand the underlying concept. Keep practicing problems like these, and you'll master the concepts related to relations and functions! You've got this!

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Here's a breakdown of how to solve it:

Understanding Equivalence Relations

First, let's recap what an equivalence relation is. A relation on a set A is an equivalence relation if it satisfies three properties:

- **Reflexive:** For every element `a` in A, `(a, a)` is in the relation.
- **Symmetric:** If `(a, b)` is in the relation, then `(b, a)` is also in the relation.
- **Transitive:** If `(a, b)` and `(b, c)` are in the relation, then `(a, c)` is also in the relation.

Equivalence relations partition the set into disjoint subsets called equivalence classes.

Solving the Problem

We need to find the number of *non-empty* equivalence relations on the set A = {1, 2, 3}. It's helpful to think about how equivalence relations create partitions of the set.

Here's a systematic way to find the equivalence relations:

1. **All elements related:** The simplest equivalence relation is where *all* elements are related to each other. This means the relation contains (1,1), (2,2), (3,3), (1,2), (2,1), (1,3), (3,1), (2,3), (3,2). This corresponds to one equivalence class: {1, 2, 3}. There's only **1** such relation.

2. **Two equivalence classes:** We can partition the set into two equivalence classes. There are three ways to do this:

- {1} and {2, 3}: This means 2 and 3 are related to each other, but 1 is only related to itself. The relation would contain (1,1), (2,2), (3,3), (2,3), (3,2).
- {2} and {1, 3}: This means 1 and 3 are related to each other, but 2 is only related to itself. The relation would contain (1,1), (2,2), (3,3), (1,3), (3,1).
- {3} and {1, 2}: This means 1 and 2 are related to each other, but 3 is only related to itself. The relation would contain (1,1), (2,2), (3,3), (1,2), (2,1).

So, there are **3** such relations.

3. **Three equivalence classes:** Each element is only related to itself. This means the relation contains only (1,1), (2,2), and (3,3). This

corresponds to three equivalence classes: {1}, {2}, and {3}. There's only **1** such relation.

Total Count

Adding up the number of equivalence relations from each case: 1 (one class) + 3 (two classes) + 1 (three classes) = 5

Therefore, the number of non-empty equivalence relations on the set {1, 2, 3} is 5. So the correct answer is (3).

Alternative method: Using Bell Numbers

Bell numbers count the number of partitions of a set. The *n*th Bell number, *B_n*, represents the number of ways to partition a set with *n* elements.

- B₀ = 1 (empty set)
- B₁ = 1
- B₂ = 2
- B₃ = 5
- B₄ = 15

...and so on.

In our case, we have a set with 3 elements, so we need to find B₃, which is 5.

Practice Problems & NCERT/PYQ References

1. **NCERT:** Review the chapter on Relations and Functions in your Class 11 NCERT textbook. Pay close attention to the definitions of relations, equivalence relations, and partitions of sets.

2. **PYQs:** Solve previous year JEE questions related to relations and functions, focusing on problems involving equivalence relations. This will help you understand the types of questions asked and the level of difficulty.

3. **Practice Problems:**

- Find the number of equivalence relations on the set {1, 2, 3, 4}.
- Let R be a relation on the set of integers Z defined by (a, b) ∈ R if a - b is divisible by 5. Show that R is an equivalence relation. Find the equivalence classes.

Key Takeaways & Motivation

- Understanding the *definition* of an equivalence relation is crucial.
- Thinking about how equivalence relations partition a set makes the problem easier to visualize.
- Bell numbers provide a shortcut for counting partitions, but it's important to understand the underlying concept.

Keep practicing problems like these, and you'll master the concepts related to relations and functions! You've got this!

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